

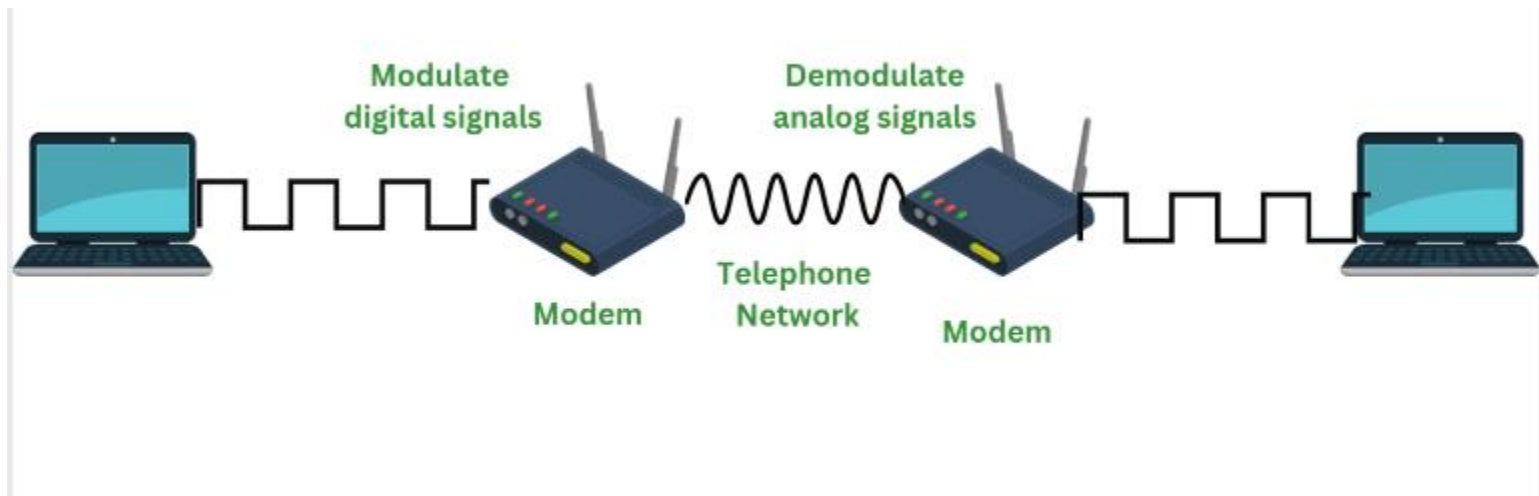
Networking Devices: Modem, Router, Switch, Access Point, Network Card

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Modem

- Converts digital signals to analog and vice versa for Internet connectivity.
- Used for connecting local networks to ISPs (Internet Service providers)



Router

- Routes data packets between different networks.
- Provides Wi-Fi and security through firewall and DHCP functions.



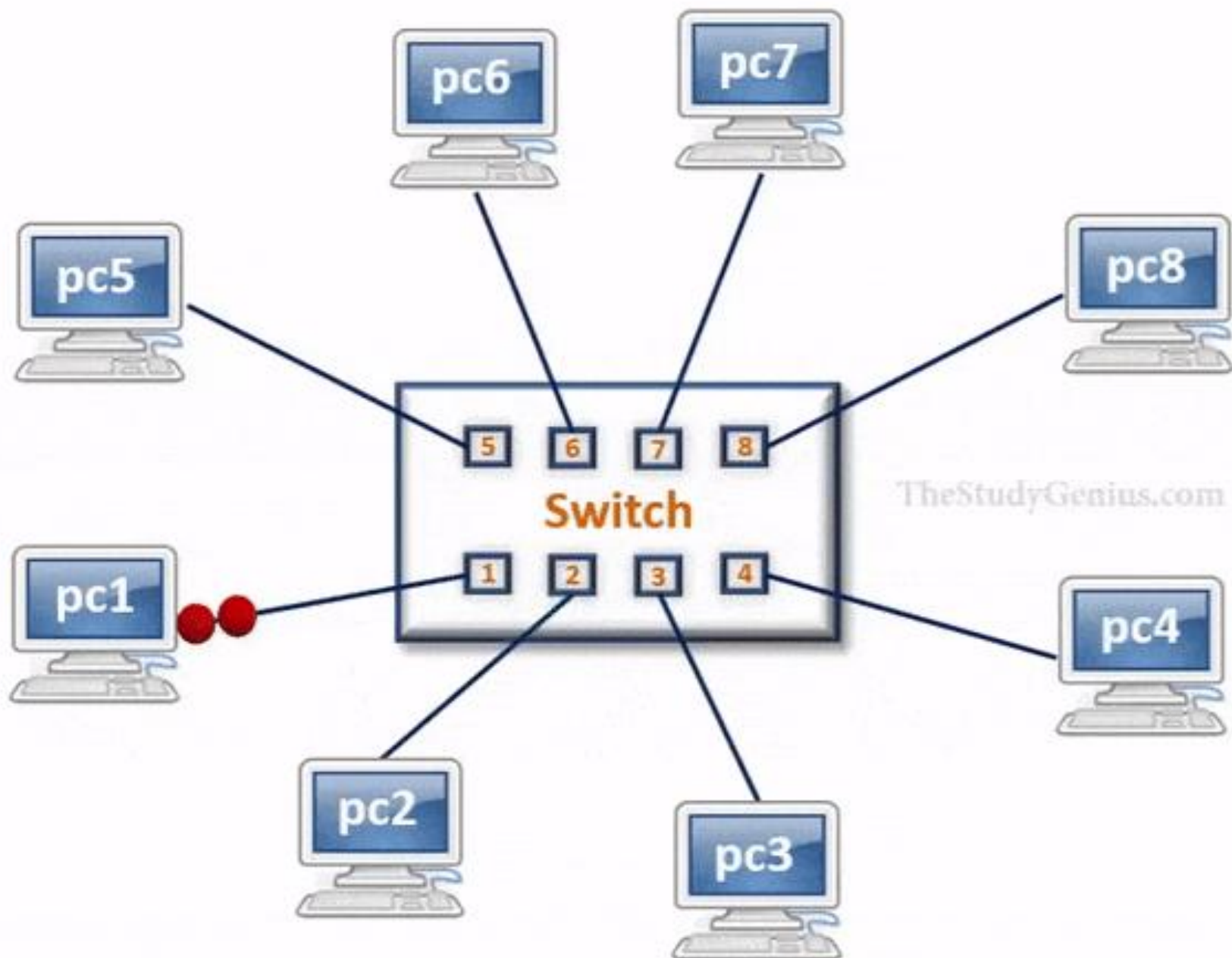
DHCP, or Dynamic Host Configuration Protocol (Additional reading)

- is a network management protocol that automatically assigns IP addresses and other configuration settings to devices on a network, like computers and smartphones. It is used to avoid manual, time-consuming, and error-prone network configuration, and to prevent IP address conflicts by ensuring each device has a unique address. T

Switch

- Connects multiple devices within the same LAN and forwards data based on MAC addresses to intended recipient.
- Uni-cast communication.
- Separate path for sending and receiving.
- Expensive and use in large environment.





Hub

- central device that connects multiple computers and other devices in a local area network (LAN).
- When a data packet arrives at one port, the hub simply broadcasts it to all other connected ports, regardless of the intended recipient.
- less efficient and less intelligent than modern network switches, which can send data directly to the correct destination.



Hub



Access Point

Access Point



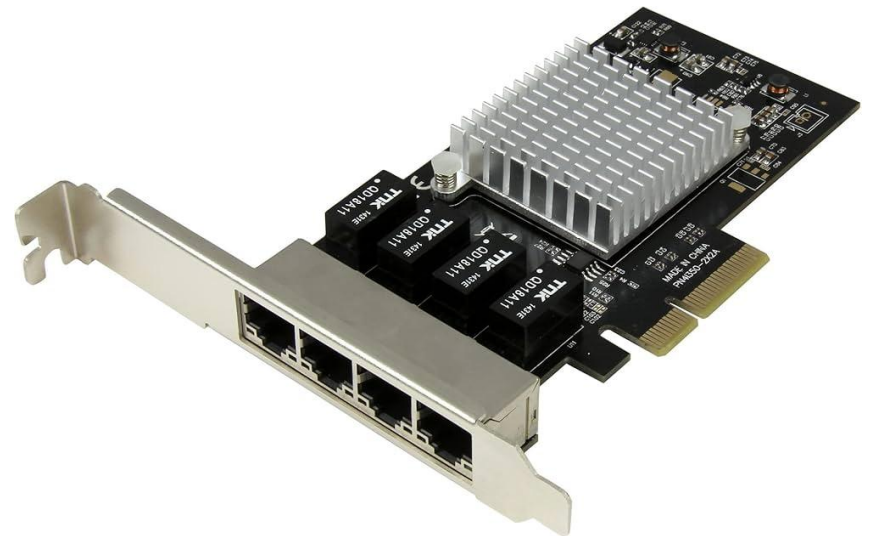
Access Point

- Allows wireless devices to connect to a wired network, extending network coverage and mobility.



Network Card (NIC)

- A hardware component that enables a device to connect to a network.
- Provides a unique MAC address for identification.



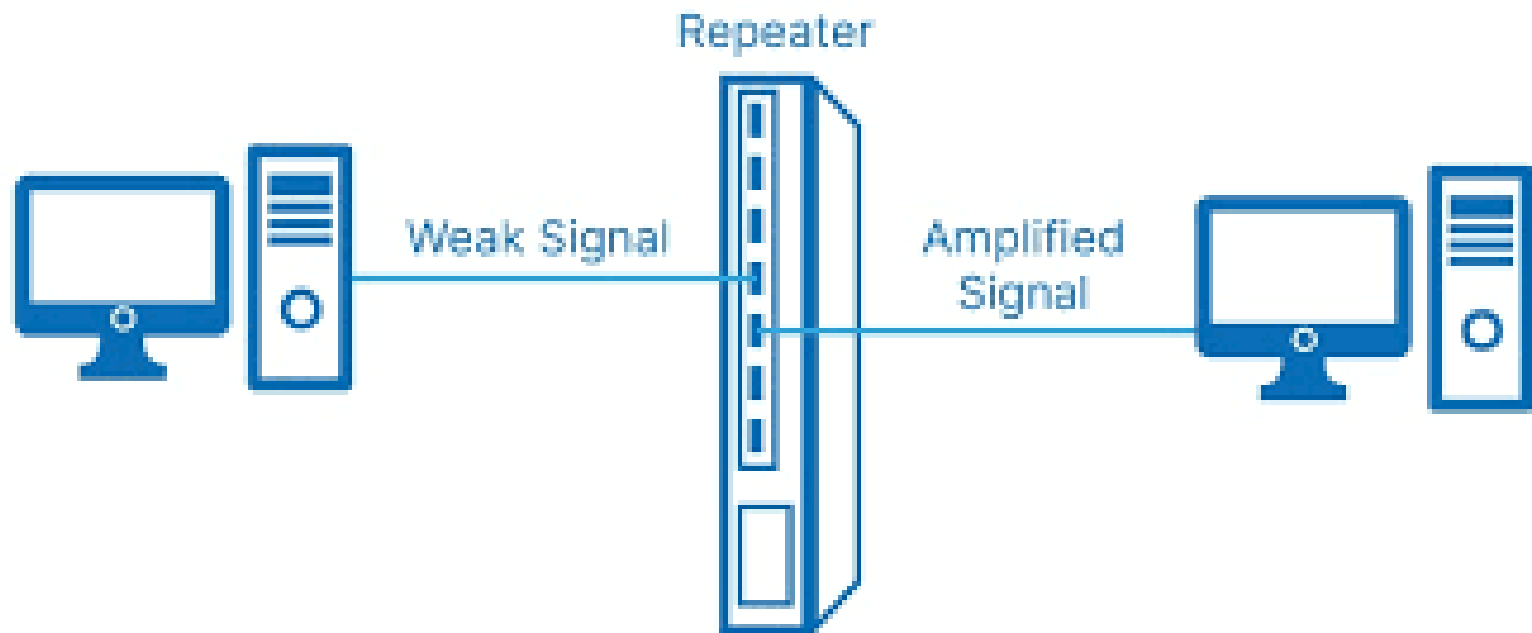
Additional reading on NIC

- A network interface card (NIC), also known as a network adapter, is a hardware component that allows a computer or other device to connect to a network. It acts as an interface between the device and the network, converting data into signals that can be transmitted over a wired or wireless network and vice versa. Most modern computers have a NIC built into the motherboard, but they can also be expansion cards, USB adapters, or integrated into other network devices.

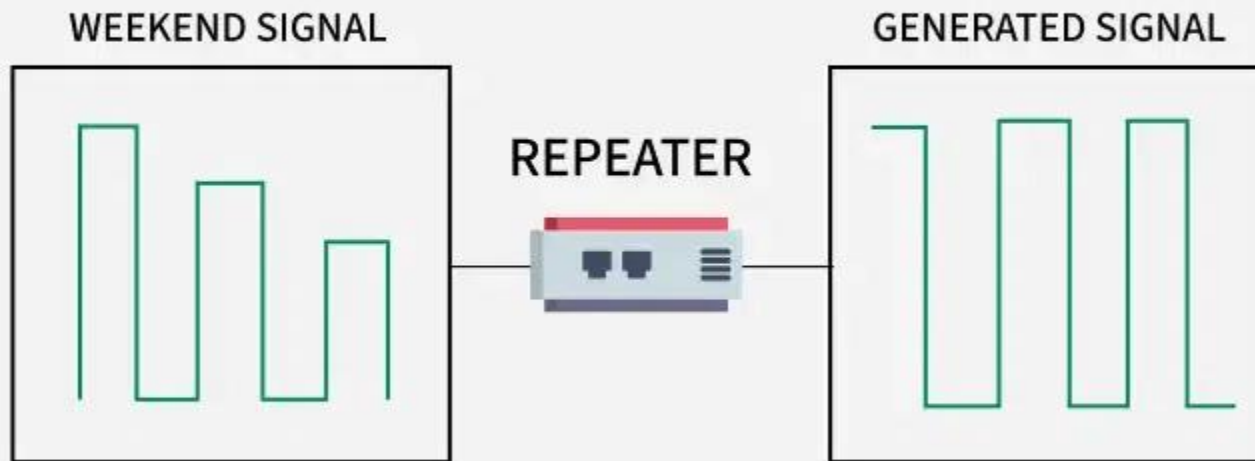
Difference between MAC Address and IP Address

- Research assignment (Ungraded)

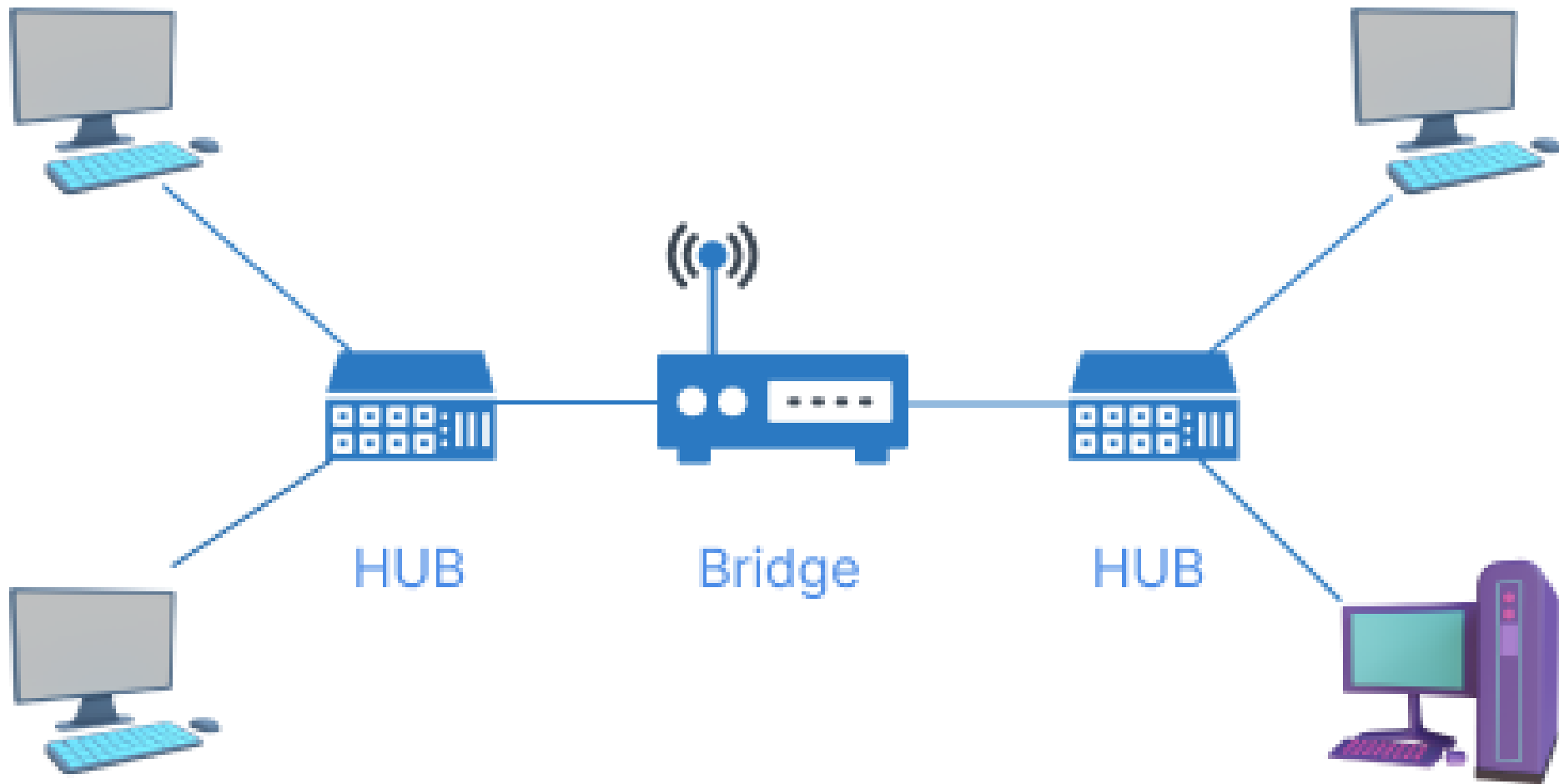
Repeater



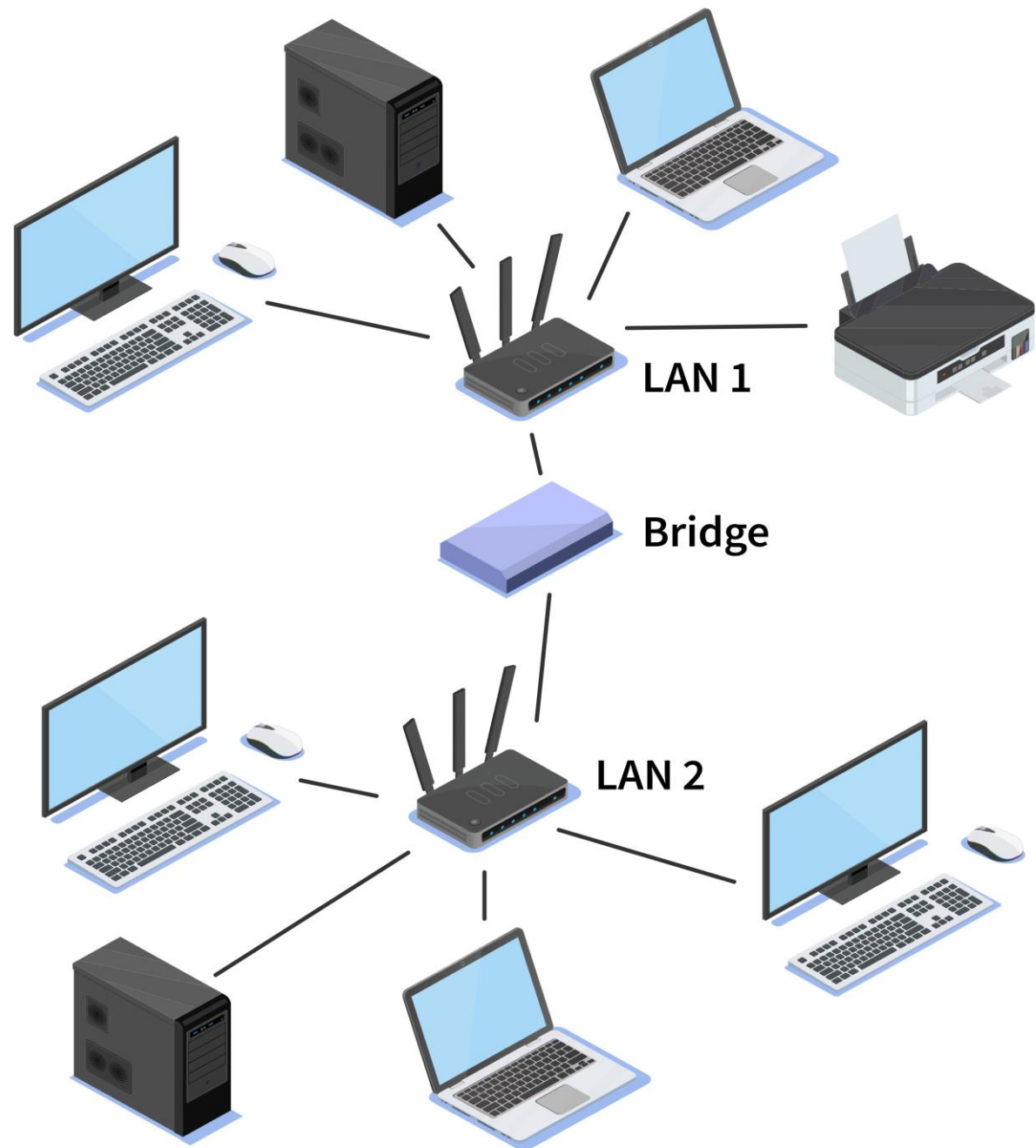
Class activity: Make definition of a repeater



Bridge



Bridge



Bridge

- A computer network bridge connects two or more network segments, allowing them to function as a single network

Summary Table

Device	Function	Example
Modem	Converts digital ↔ analog signals for Internet connectivity	DSL Modem
Router	Routes data between different networks	Wi-Fi Router
Switch	Connects multiple devices in a LAN	Ethernet Switch
Access Point	Provides wireless access to a wired network	Wi-Fi Access Point
Network Card (NIC)	Enables a device to connect to a network	NIC / Wi-Fi Adapter



Modem



NIC



Repeater



Hub



Switch



Router



Bridge



Gateway

Conclusion

- Each networking device plays a key role in ensuring connectivity and communication.
- Understanding their functions helps in network design and troubleshooting.

Q&A / Discussion

- • How is a router different from a switch?
- • Which devices are needed for a small office network?
- • Can a single device perform multiple roles?